

HomeWiFiNetwork

1) Create a home Wi-Fi network in Cisco Packet Tracer using three PCs and one wireless router, and save the project as 'HomeWiFiNetwork.pkt'.

Step 1: Open Cisco Packet Tracer

- Open Cisco Packet Tracer on your computer.
- Create a new project.
- Save the project as HomeWiFiNetwork.pkt.

Step 2: Add Network Devices

- From End Devices, add three PCs to the workspace.
- From Network Devices / Wireless Devices, add one wireless router.
- Arrange the devices neatly on the workspace.

Step 3: Connect the PCs

- Use Copper Straight-Through cables.
- Connect PC0 FastEthernet0 to a LAN/Ethernet port on the router.
- Connect PC1 FastEthernet0 to another LAN/Ethernet port.
- Connect PC2 FastEthernet0 to a third LAN/Ethernet port.
- Use the router's LAN ports, not the Internet/WAN port.

Screenshot to Include

- Completed Packet Tracer topology showing three PCs and the wireless router.

2) Assign static IP addresses to each PC so that all devices are on the same subnet and configure the router's IP as the default gateway.

Step 1: Configure the Router

- Open the wireless router configuration.
- Set the LAN/local IP address to 192.168.10.1.
- Set the subnet mask to 255.255.255.0.
- The router address 192.168.10.1 will be used as the default gateway.

Step 2: Configure PC0

- Open PC0 → Desktop → IP Configuration.
- Select Static.
- IP Address: 192.168.10.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Step 3: Configure PC1

- Open PC1 → Desktop → IP Configuration.
- Select Static.
- IP Address: 192.168.10.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Step 4: Configure PC2

- Open PC2 → Desktop → IP Configuration.
- Select Static.
- IP Address: 192.168.10.4
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

IP Addressing Summary

- Router: 192.168.10.1
- PC0: 192.168.10.2
- PC1: 192.168.10.3
- PC2: 192.168.10.4
- Subnet Mask: 255.255.255.0
- Network: 192.168.10.0/24

Screenshot to Include

- PC0 IP Configuration window.
- PC1 IP Configuration window.
- PC2 IP Configuration window.

3) Test connectivity using the Packet Tracer ping tool and verify that all PCs can communicate with each other and the router.

Step 1: Open Command Prompt

- Open PC0 → Desktop → Command Prompt.
- Use the ping command to test PC1, PC2, and the router.

Step 2: Test from PC0

- ping 192.168.10.3
- ping 192.168.10.4
- ping 192.168.10.1

Step 3: Test from PC1

- ping 192.168.10.2
- ping 192.168.10.4
- ping 192.168.10.1

Step 4: Test from PC2

- ping 192.168.10.2
- ping 192.168.10.3
- ping 192.168.10.1

Expected Result

- Each ping should receive a Reply from the destination.
- A successful test should show 0% packet loss.
- If the first ping fails, repeat the command after a short delay.

Screenshot to Include

- Command Prompt showing successful ping replies between the PCs.
- Command Prompt showing a successful ping to the router.

4) Troubleshoot connectivity if any PC cannot communicate with another device.

Step 1: Check IP Configuration

- Confirm every PC has a unique IP address.
- Confirm all PCs use subnet mask 255.255.255.0.
- Confirm all PCs use 192.168.10.1 as the default gateway.

Step 2: Check Physical Connections

- Make sure each PC is connected to a LAN port on the router.
- Check that the Ethernet links become active.
- Check the cable type and selected ports.

Step 3: Check for IP Address Errors

- Make sure there are no duplicate IP addresses.
- Check that each IP address is in the 192.168.10.0/24 network.
- Correct any typing mistakes.

Step 4: Repeat Connectivity Test

- Open Command Prompt on the affected PC.
- Run ping 192.168.10.1 to test the gateway.
- Run ping to the other PC's IP address.
- If the gateway responds but another PC does not, check the other PC's configuration and connection.

Troubleshooting Summary

- Problem: PC cannot reach another PC.
- Cause: Incorrect IP, subnet mask, cable, or port.
- Resolution: Verify the IP configuration and physical connection.
- Problem: PC cannot reach the router.
- Cause: Incorrect default gateway or router LAN configuration.
- Resolution: Verify the router address is 192.168.10.1 and the PC gateway is correct.